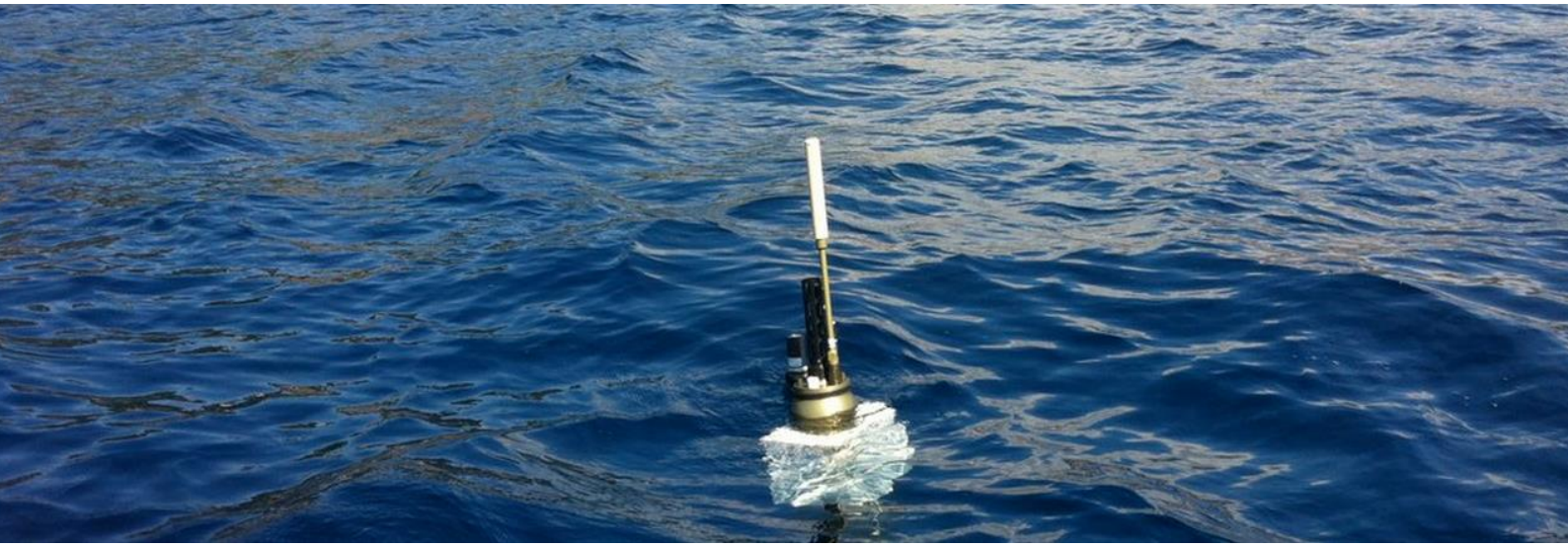
# PFV2 Profiler – Under Ice Capabilities

**P**ROFILING **F**LOAT **V**ECTOR **2**



NT000075_PFV2_Under_Ice_Capabilities
Revision 1.0 (2025-02-18)

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# 1. Revision history

| Revision | Release date | Notes | Author |
|---|---|---|---|
| 1.0 | 2025-01-22 | Draft | D. NOGRE |
| | | | |

# 2. Introduction

The use of profiling floats in the polar oceans is seriously impeded by the presence of sea ice as floats are not able to transmit their data. An ice-avoidance algorithm is essential for ice floats.

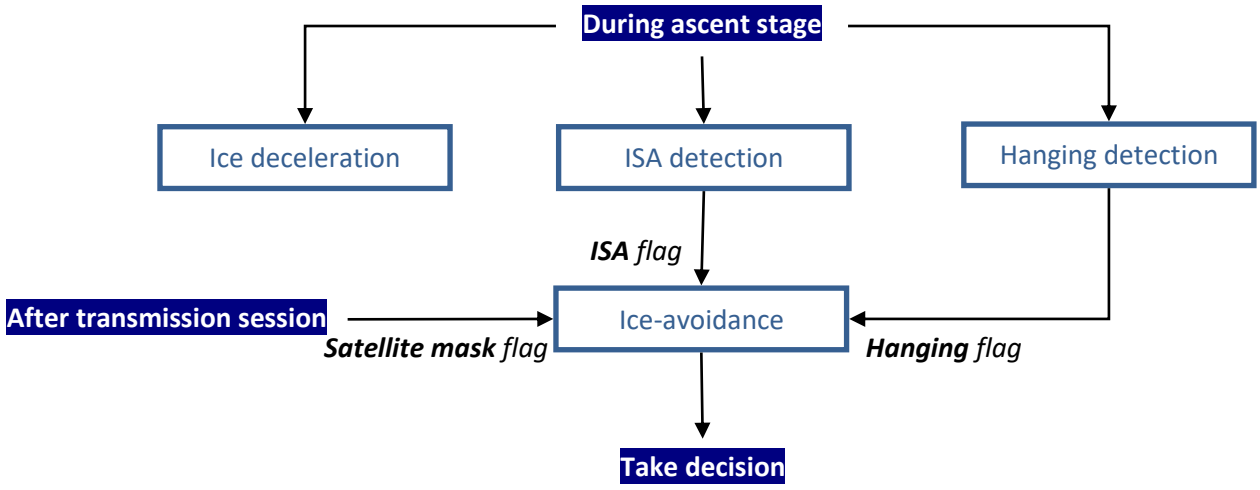
The main objective of ice-avoidance algorithm is to prevent the float from being damaged by knocking the underside of the sea ice or crushing in free-floating ice blocks at sea surface. The second objective is to save energy when surfacing is not efficient.

# 3. Under-ice features

## 3.1 Overview

To safely navigate and collect data under the ice, ice floats need specific features compared to floats operating in open waters:

- Reduction of the ascent speed when approaching the surface
- Mechanism for detecting ice cover:
    - Ice-sensing algorithm prediction
    - Hitting the underside of the sea ice
- Evasion mechanism to abort surfacing
- Ability to store and keep in memory important amount of data

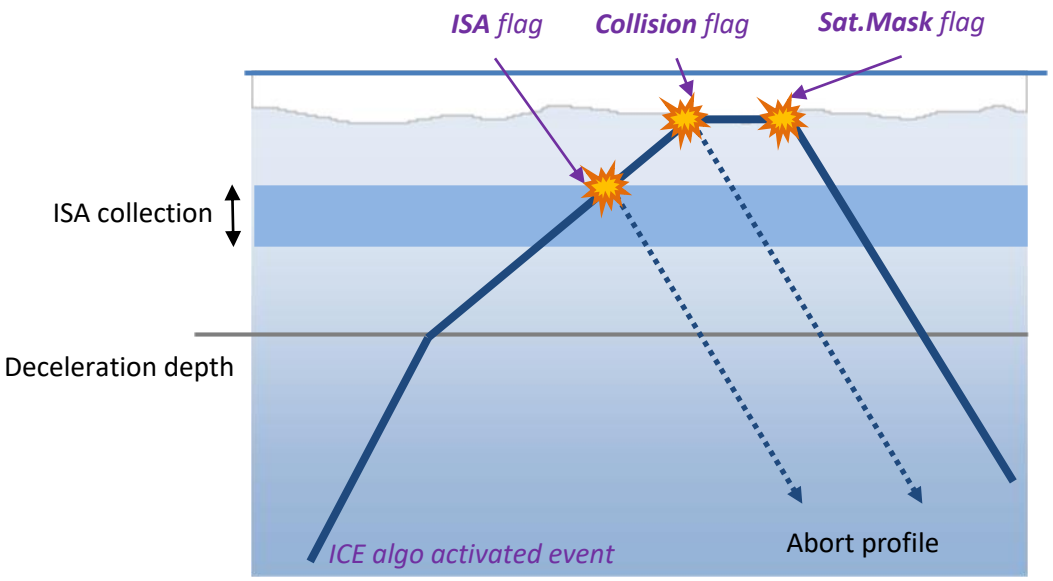



**--- Figure 1. Under-ice features ---**

The final decision considers different states of the float:

- **ISA** flag based on the result of the ice-sensing algorithm
- **Hanging** flag if the float is blocked in the underside of the sea ice
- **Satellite mask** flag exploits the Iridium system to determine if the float is at the surface with sky view

When the algorithm takes the decision to abort a profile, technical and sensor data are stored in the memory. The float keeps the data collected into its memory until transmission.




**--- Figure 2. Under-ice decision ---**

## 3.2 Ice option

The ice detection mechanism is made possible by a firmware option. This option can be requested when ordering the product but can also be sent by Iridium download if the float has already been deployed.
To enter the option key, please enter the command (from the GUI or by generating an Iridium remote control file): **" !option-key:x y"**
Where:

- "x" is the option number (0 for ICE option)
- "y" is the option key (the key is an 8-character hexadecimal number)

When the Ice option is valid, the ice parameters section and the ice "pictogram" are visible. And the XML setting file will include the key at <SETTING-ICE-P0>.

## 3.3 Ice activation

The ice activation is conditioned from 2 parameters (also available from the GUI):

- A global activation: <SETTING-ICE-P1>

**AND**

- A calendar activation: <SETTING-ICE-P2> - The parameter triggers the ice algorithm when the current date is one of the selected months.
  *(example: 110000000001 means that ice detection is used from December to February).*

Once the 2 conditions joined, the mechanism of ice avoidance will be launch at the beginning of the ascent phase.

In parallel, all securities of the ice avoidance can be deactivated separately:

- ISA algo.: <SETTING-ICE-ISA-P0>
- Hanging: <SETTING-ICE-HANGING-P0>
- Satellite masking.: <SETTING-ICE-SAT_MASK-0>

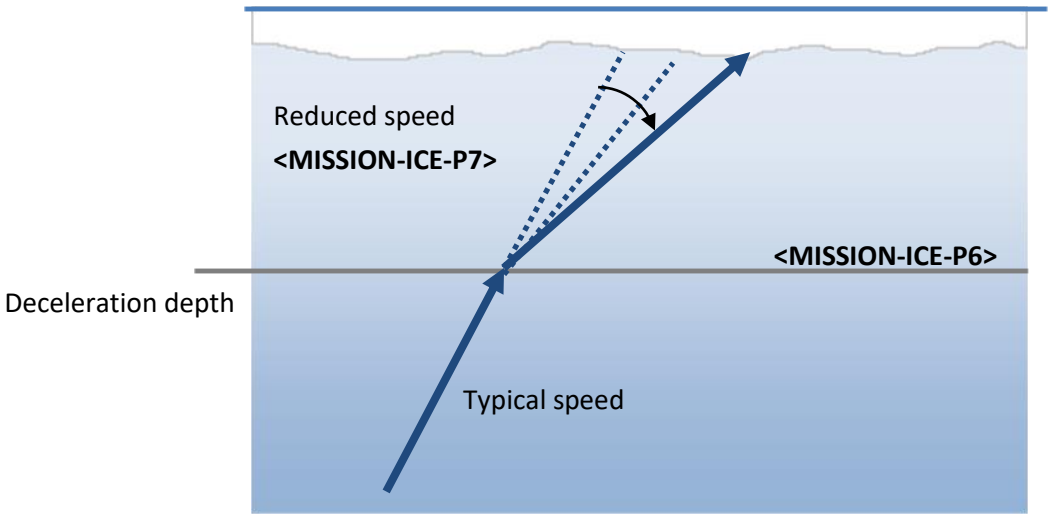
## 3.4 Ice deceleration

This feature consists in dividing the ascent in different stages in order to allow the float to detect the presence of ice and to abort the profile.

The operation is as follows:

- Stage 1: "Standard" ascent to the deceleration depth
- Stage 2: "Slow" ascent to surface

The “deceleration power” can be adjusted to obtain better results. More the ascent speed is reduced, more the ascent stage is extended.



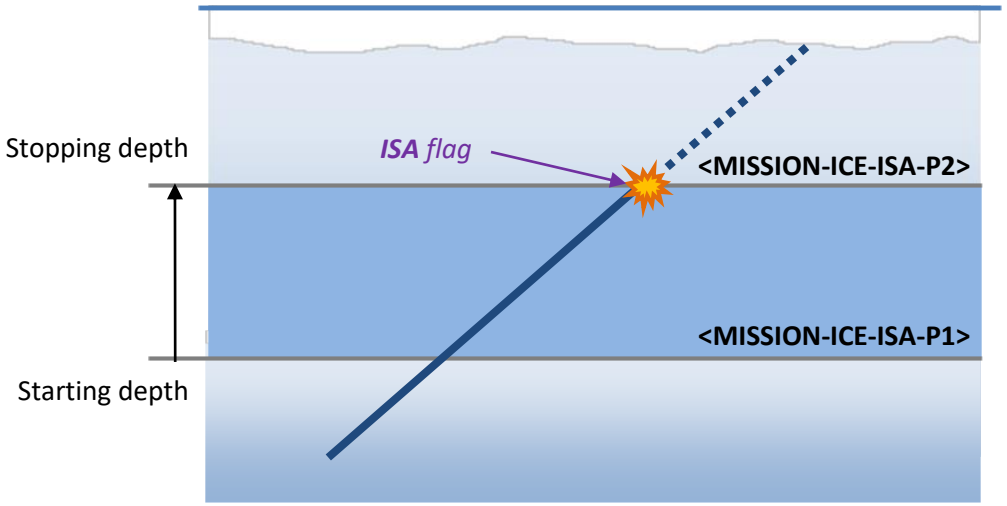

**--- Figure 3. Ice deceleration ---**

The reduced speed **<MISSION-ICE-P7>** is limited by the minimum and maximum speed ascent defined in technical node (**<TECHNICAL-ASCENT-MIN-P0>** / **<TECHNICAL-ASCENT-MAX-P0>**).

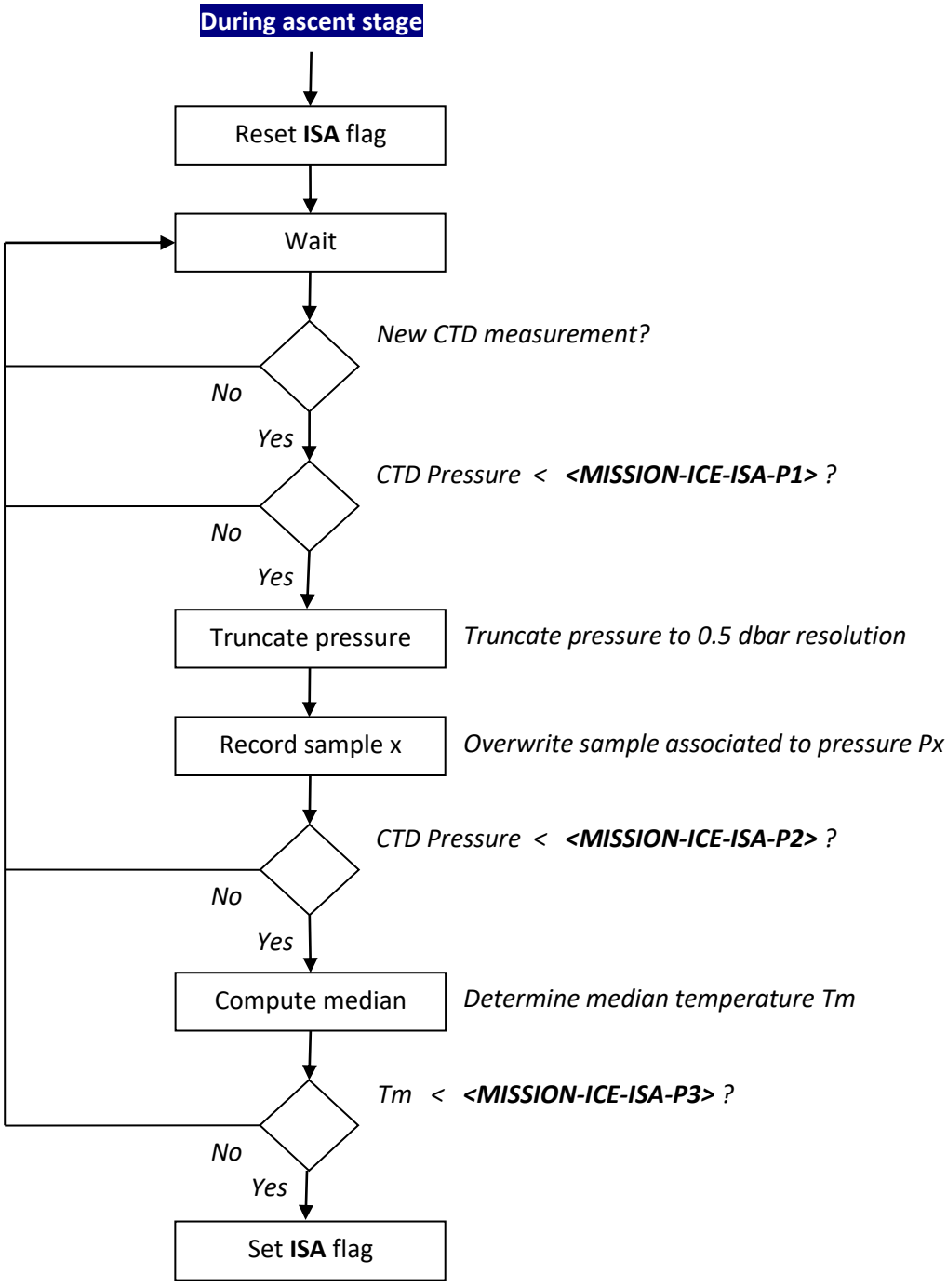
## 3.5 Detect ISA: ice-sensing algorithm (ISA)

This algorithm relates subsurface water temperatures to the probability of sea ice at the surface.

Temperature samples are collected during ascent stage, sorted, and stored every 0.5 dbar. The algorithm calculates the median temperature within a pressure interval close to the surface and compares it with a predefined threshold.



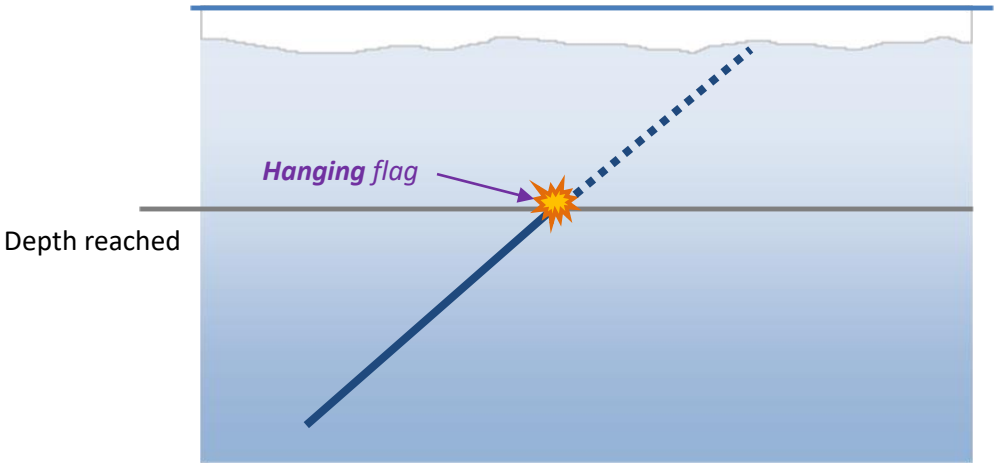

**--- Figure 4. Detect ice by ice-sensing algorithm ---**




**--- Figure 5. Ice-sensing algorithm ---**

## 3.6 Detect hanging: hitting the underside of the sea ice

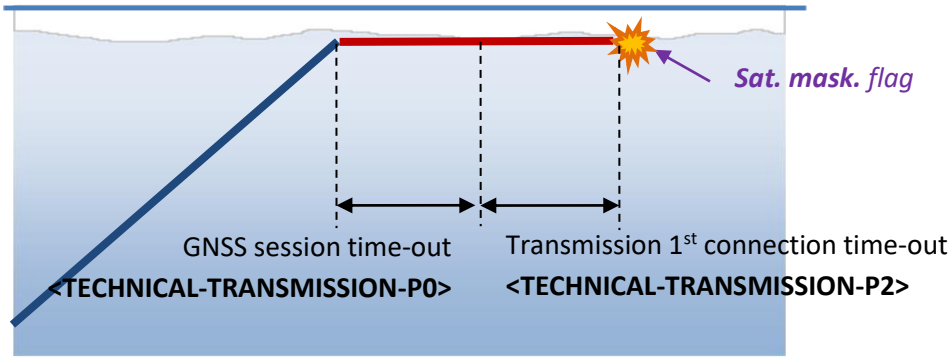
The detection of the underside of the sea ice is made by the hanging detection feature of the float. It consists in detecting the absence of movement of the float despite a large displacement of oil to ascend.




**--- Figure 6. Detect ice by hitting the underside of the sea ice ---**

## 3.7 Detect satellite masking: no satellite view

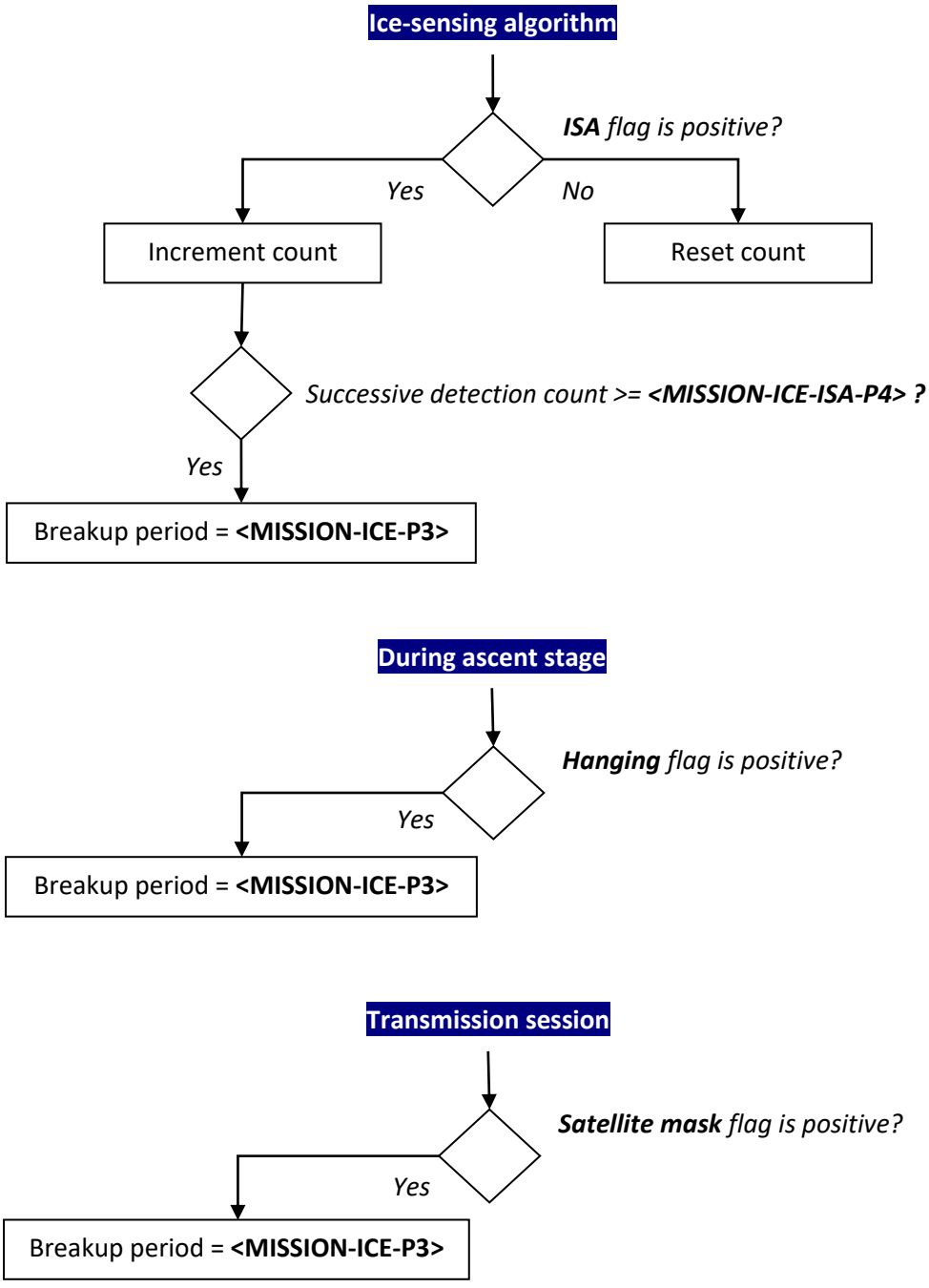
The coverage flag, indicating that the float has no view of the satellites, is positioned when the float cannot connect to the satellites (GNSS ***AND*** transmission).




**--- Figure 7. Detect ice when no sky view ---**

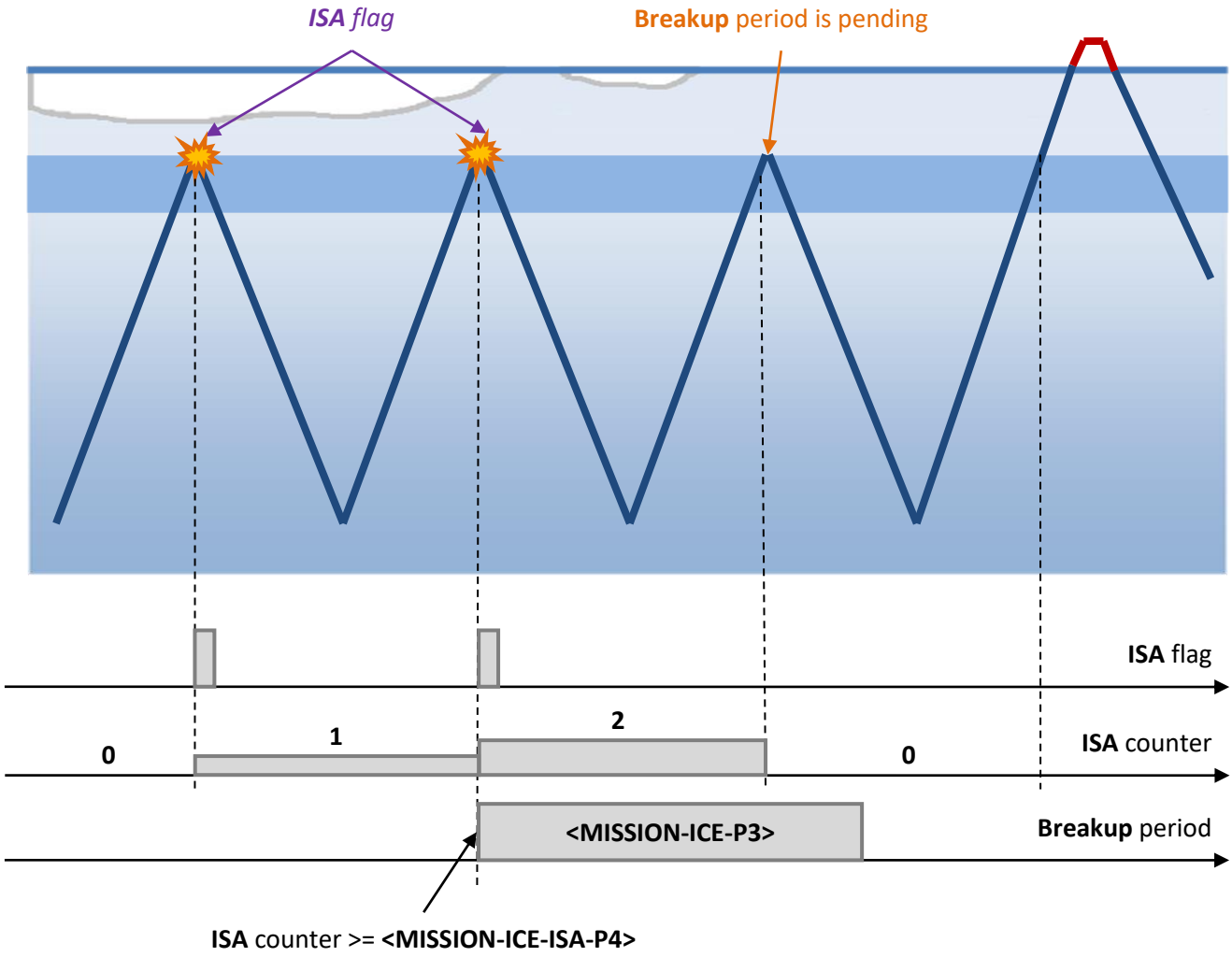
## 3.8 Detect ice: maintaining period – persistence mechanism

The idea behind the ice detection maintaining period is to postpone surface session for a predetermined number of days after the last ice detection. This delay allows time for the ice to break-up and the profiling float to find a new ice-free zone.




**--- Figure 8. ICE period management ---**

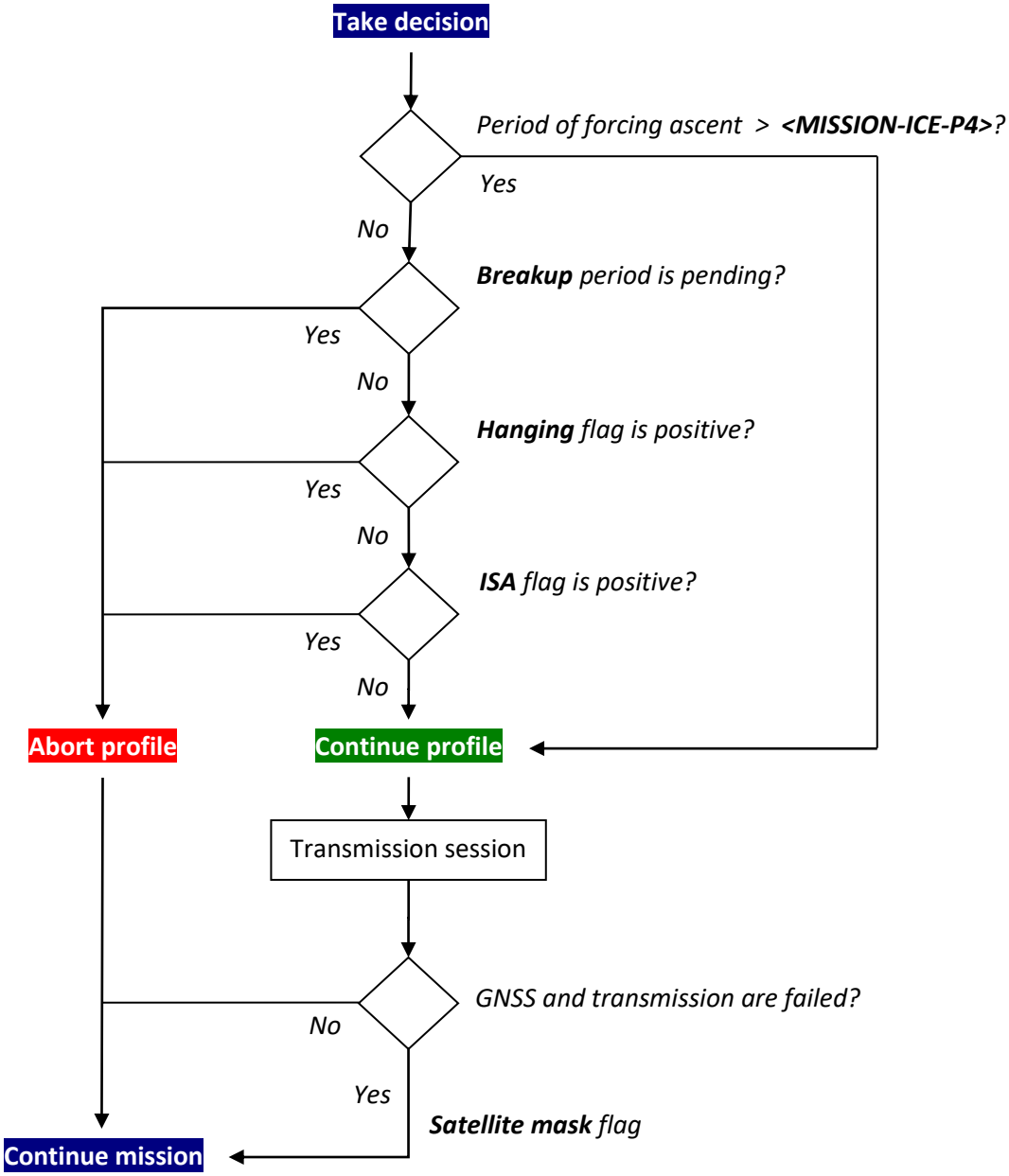
In the event that ice is detected by the ISA algorithm, a successive ISA detection counter is incremented to filter the information.




**--- Figure 9. ISA counter vs. Breakup period ---**

**Note:** In case the ice-sensing algorithm is aborted by collision detection, the ISA flag is not set and the successive ISA detection counter is not changed.

## 3.9 Take decision: ice-avoidance algorithm

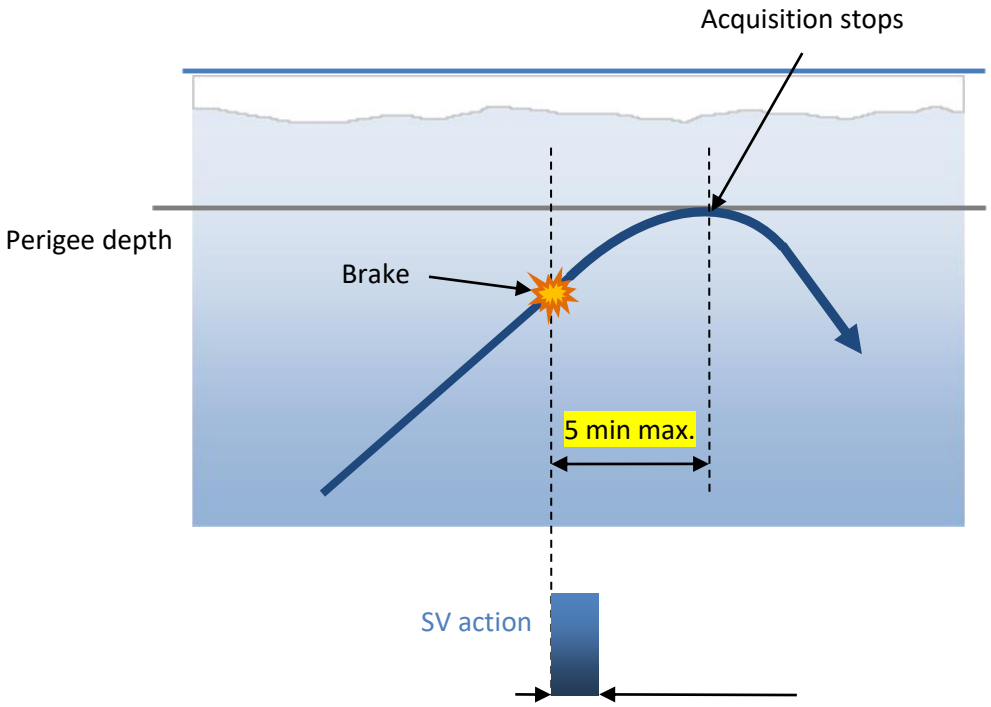



**--- Figure 10. Ice-avoidance algorithm ---**

## 3.10 Aborting a profile

Once the decision to abort a profile is taken, the float immediately reduces its buoyancy by braking.

The acquisition remains active until the detection of the ascent + (5 minutes time-out since the brake action if no buoyancy inversion observed).




**--- Figure 11. Aborting profile ---**

# 4. Parameter setting

Please refer to the documentation **"NT000068_PFV2_Parameters_Set".**

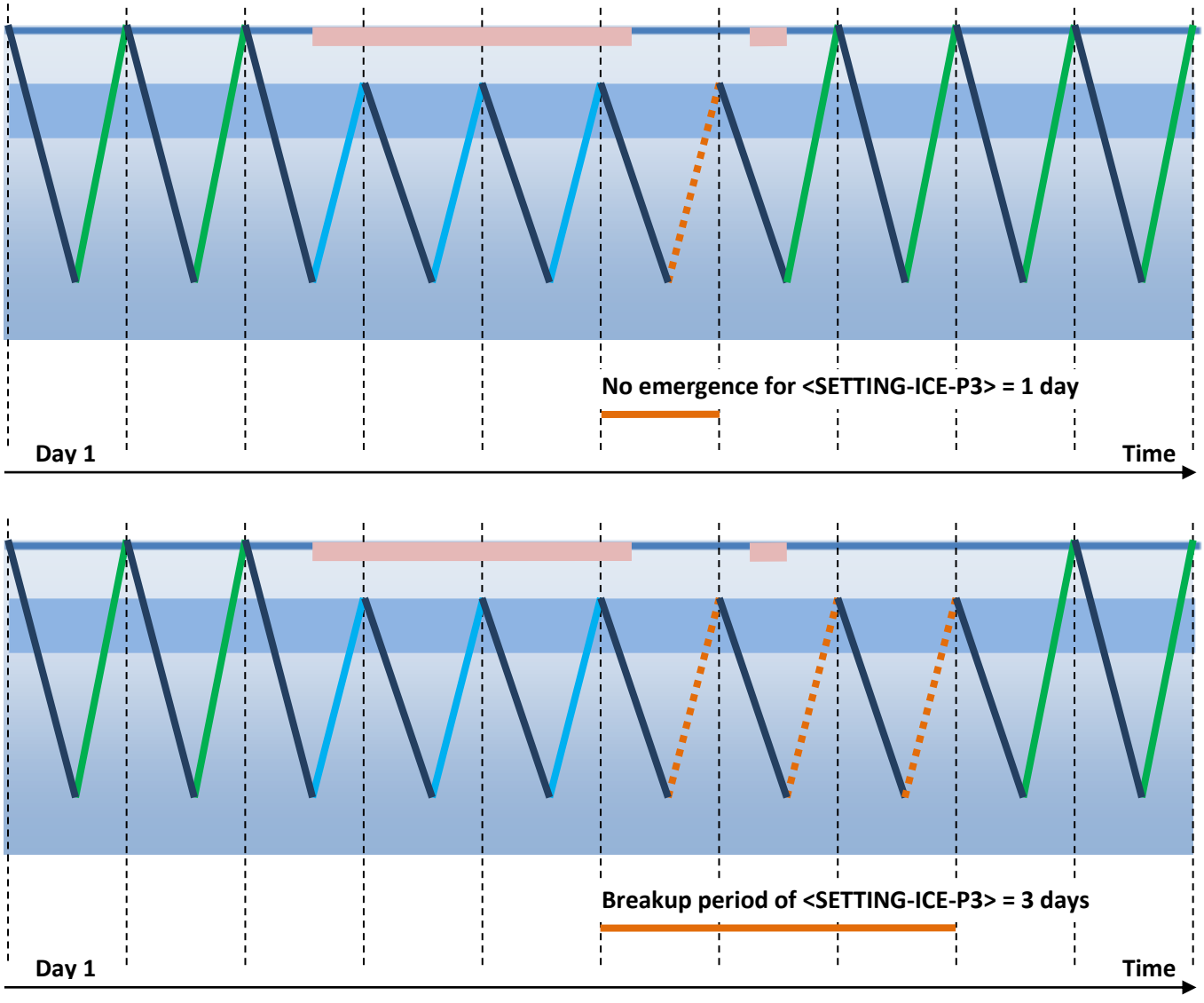
# 5. Appendices

Examples below show under ice mechanisms and impacts of different settings applied to a 1 day profile mission:

- Represents the presence of sea ice at the surface
- Represents an ascent stage which finishes with a surface session (transmission success)
- Represents an ascent stage which finishes with an ice collision
- Represents an ascent stage aborted by ice detection mechanism (ISA flag)
- Represents an ascent stage aborted by ice persistence mechanism (Breakup period)

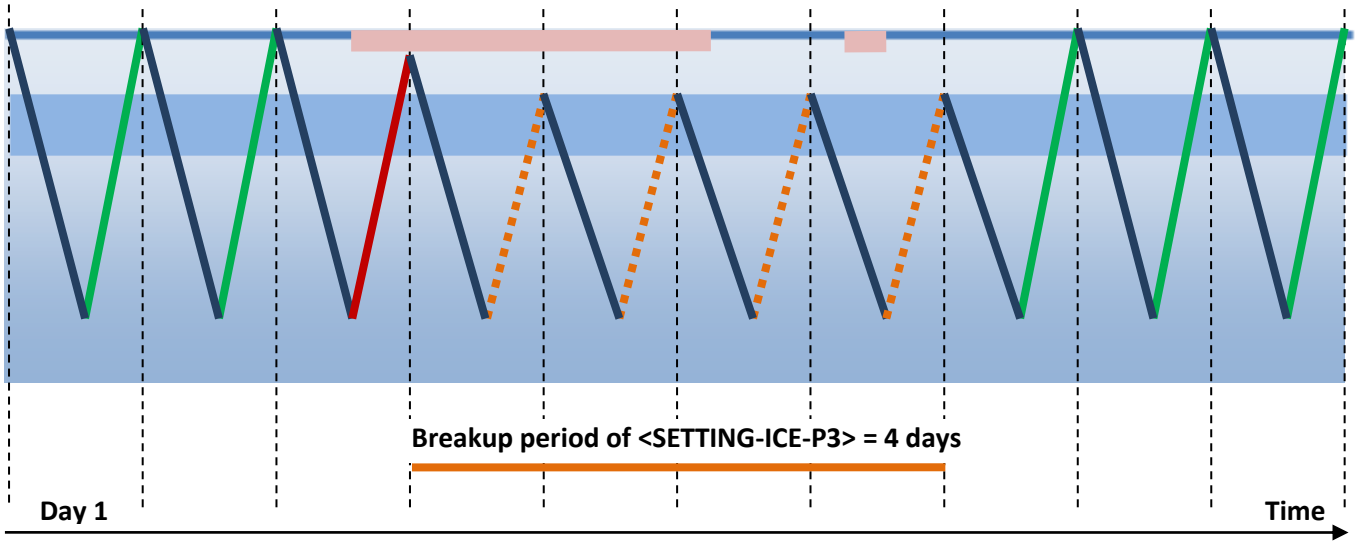
## 5.1 Example 1: <SETTING-ICE-P3> parameter - on ISA detection

This parameter is used to maintain the ISA flag positive after the last detection has occurred.



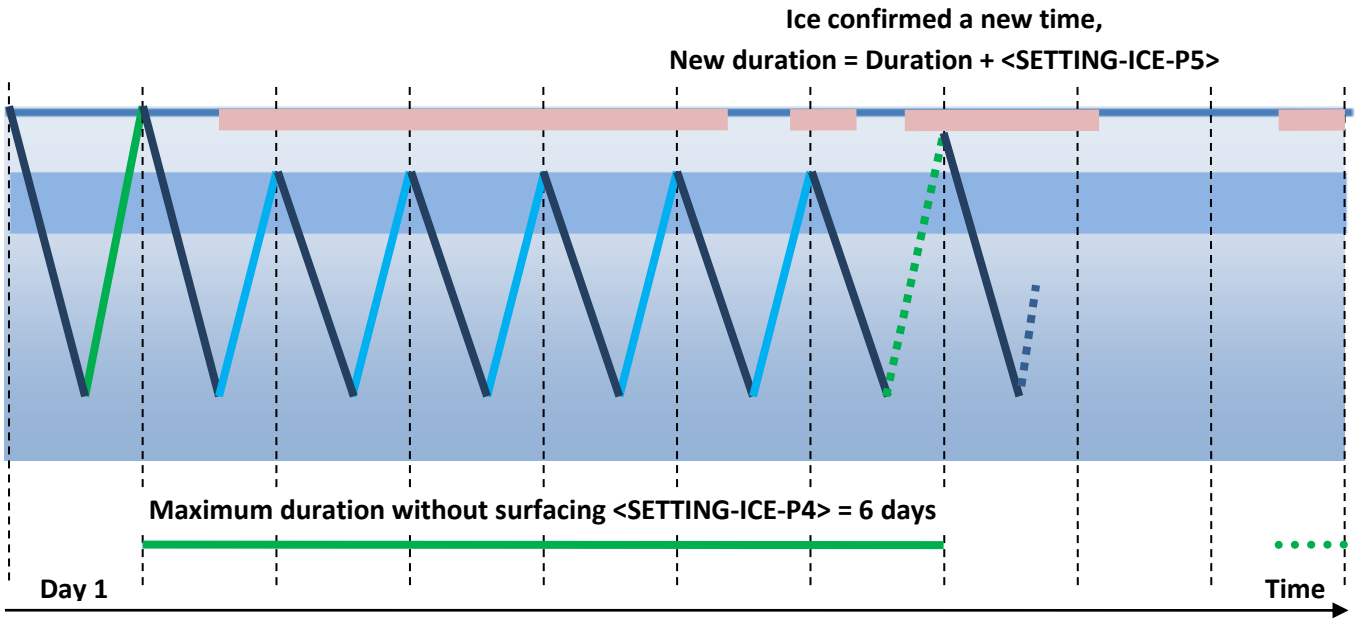
## 5.2 Example 2: <MISSION-ICE-P3> parameter - on Hanging or Satellite mask detection

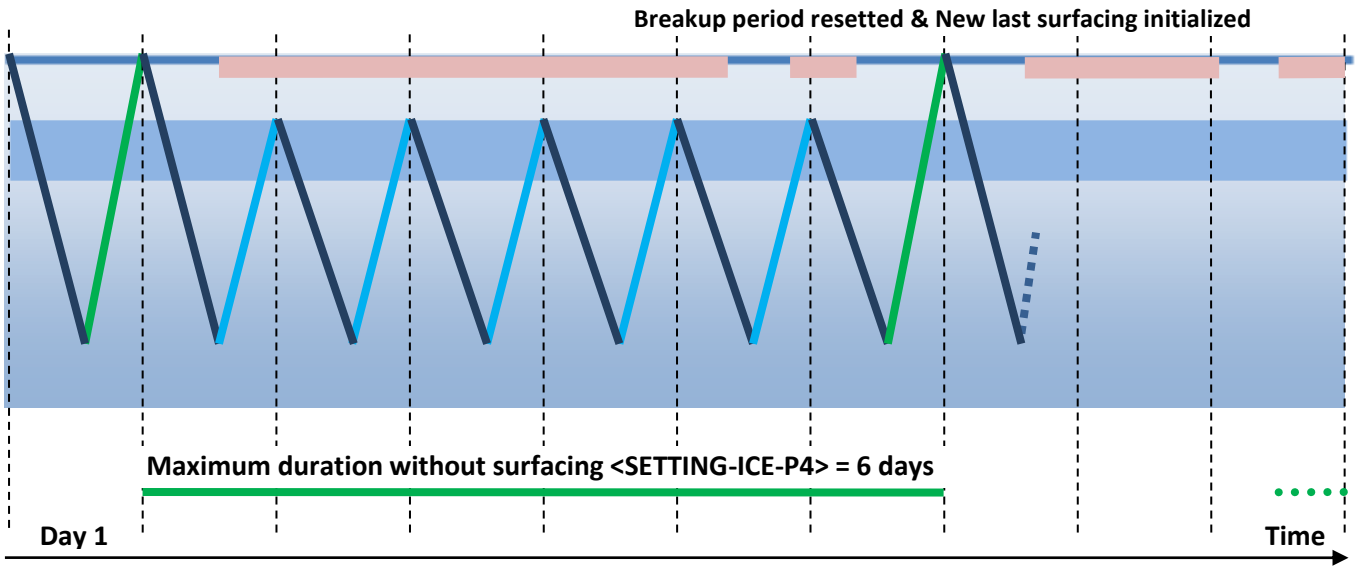
This parameter is used to maintain the hanging (or satellite mask) flag positive after the last detection has occurred.




## 5.3 Example 3: <SETTING-ICE-P4> parameter

This parameter is used to force a surface session after a while without surfacing (even if ISA flag is positive or ICE period is in progress). The duration counter is restarted after an effective surfacing (with transmission success).

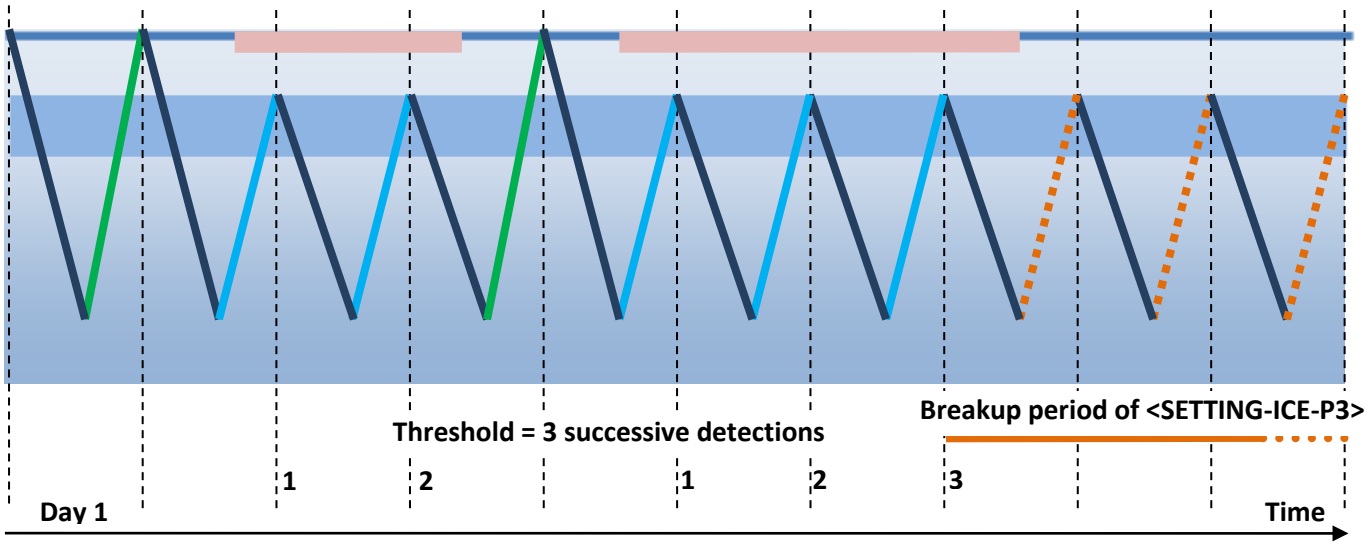



Breakup period resetted & New last surfacing initialized
Maximum duration without surfacing <SETTING-ICE-P4> = 6 days
Day 1
Time

## 5.4 Example 4: <SETTING-ICE-ISA-P4> parameter

This parameter is used to wait for successive ISA detections before applying the breakup period (prohibition on emergence) of <SETTING-ICE-P3> days.

Example: If <SETTING-ICE-ISA-P4> = 3



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